The University of Oxford

MSc (Mathematics and Foundations of Computer Science)

Analytic Topology

Hilary Term 2026

This mini project is not a sequence of tasks you are asked to carry out, but rather a discussion and exploration of a mathematical structure similar to topological spaces. As such it is very open ended and gives you the opportunity to put your own ideas forward and make judgement calls as to what is interesting to investigate and present.

As a rough guide the written text of your mini-project should be roughly 10 pages in length. If you use pictures or diagrams to illustrate your text, these should be considered as extra.

A σ-topology on a set X is a collection of subsets of X that is closed under binary intersections and countable unions.

Your task is to discuss σ-topologies and compare them with standard topologies. Illustrate you discussion with concrete examples where appropriate. Keep in mind that characterizations of properties or constructions that are equivalent in the setting of usual topologies may not be equivalent for σ-topologies any more.

Below is a suggestion of topics you could cover. This list is neither exhaustive, nor is it required to cover all of these. You should feel free to focus your attention on some of these which you cover in greater depth. I have also grouped the topics together, but again you should feel free to discuss them in a different order. Although your mini-project is necessarily linear, you should feel free to refer forwards where it helps the understanding or motivation.

If proofs or parts of proofs from standard topology can be used (e.g. from the lecture notes or standard reference works) there is no need to repeat these and referencing them is sufficient.

Motivation

Explain in the context of the real line and a computational model of the reals why open sets can be interpreted as those subsets A of $\mathbf{R}$ such that if a real number x belongs to A then this can be 'decided'. With this interpretation of open sets or otherwise, motivate the study of σ-topologies.

You are encouraged to comment on this computational viewpoint in the remainder of the mini-project. If you would like to use some non-classical (constructive) logic instead of the traditional first order logic, you are free to do so formally or comment on it informally, but this is strictly optional.

Basic Constructions

Discuss analogues of closed sets, closures, basis, subbasis, continuity, convergence, products and subspaces.

Separation Axioms and Metrizability

Discuss analogues of the separation axioms and metrizability results for σ-topologies.

Compactness Properties

Discuss analogues of compactness properties and the interplay of compactness properties and separation axioms.

Advanced Topics

Function Spaces, compactifications, connectedness, dualities along the lines of frames, Stone duality and Priestley duality.

Examples

Discuss examples of topological spaces and their properties e.g. metric spaces (discrete spaces, topology of uniform convergence for metric-valued functions), the Sorgenfrey Line (and related spaces), linearly ordered spaces, $\beta\mathbf{N}$ and its subspaces.